

Responses to Reviewer Questions

Here are the answers to the proposed questions. In the accompanying PDF, our corrections and specific additions to the manuscript have been marked in blue for clarity.

1. I took out "long-range correlations" from the abstract. This has not been investigated and showed that there are indeed long-range spatially correlated processes. For that, one would have to do correlation analysis such as DFA and show that the correlation function is also a power law. But if you have a power law distribution, then this can create an artificial correlation just because the power law has these large events. If you reshuffle a power law distributed and correlated sequence, you still get similar correlation properties as in the sequence before reshuffling since there are large events throughout no matter how you reshuffle. This is not to say that there are no power-law correlations, but the paper did not show this specifically which is why I suggest taking out the statement. Understood. Since we did not perform that analysis, I agree we cannot use that term.
2. While DLA is a nice process for creating fibril structures, reality is different. There is no sphere around a first molecule. It's more like a well-orchestrated biological process creates a core, called a microfibril, that usually contains 5 molecules. Then the cell holds the microfibril, creates more of them and the assembly grows by the cell's active energy-dependent manipulation of where new microfibrils are placed. So it is clearly not a random process like the DLA. Nevertheless, I agree the DLA is a good model. I am not suggesting that the model should be changed. I am just saying the limitation of the DLA approach. I should also say that the fact that molecular rotation for example in the x-y plane or the y-z plane, is not considered mimics the constraints a cell would exert on fibrillogenesis.

I believe the last paragraph, which addresses the model's limitations, was too brief. I revised it to make it clearer and incorporated the point you raised about the DLA.

3. In figure 4, the backbone is black but the rest is gray, kind of dark gray and sometimes it is unclear whether an active molecule is present. Is it possible to make the gray lighter, or change the black to red or blue? It's also confusing because in figure 5, gray represents the backbone. So perhaps you could have very light gray for all molecules and dark gray for backbone which would match the dark gray in figure 5?

I modified the colors: dark blue now represents the removed portion of the molecule, gray represents the backbone, and green highlights a specific molecule used in Figure 5 to illustrate the calculation.

4. I did not notice this before, but I wonder about Eq. 4. The Weibull cumulative distribution is usually given as $F(x) = 1 - \exp[-(x/\lambda)^m]$. The density distribution will have an exponential term. Without the exponential, the ratio can be larger than 1 which cannot represent a probability. Also, is it true that the sum of $N(i) = K$? Anyway, I do not have references 35 and 39, so this might be developed in those papers. If so, this is okay; I'm just curious.

The parameter $N(i)$ is the number of elements in cross-section i , and K is the total number of elements that are nearest neighbors of a given molecule in each layer — that is, K is the

sum of all elements sharing a face with the molecule of interest. We assume that each of these K elements forms a bond with that molecule, which is why the denominator of P_R is $K\sigma_c$: the total force required to release a molecule from the active backbone.

Regarding the possibility of $P_R > 1$: you are correct that the expression $(\sigma_M/K\sigma_c)^m$ can mathematically exceed unity, particularly in the advanced fracture regime, when the number of remaining elements in the active backbone decreases and each one bears a larger load. However, this does not represent a physical inconsistency in the model: when $P_R \geq 1$, the element is removed with certainty during stochastic sampling — behavior identical to the natural limit of the full Weibull distribution, which saturates at $F \rightarrow 1$ when $x \gg \lambda$. The complete form of Weibull distribution in the terms of $x = \sigma_M$ and $\lambda = K\sigma_c$:

$$P_R = 1 - \exp \left[- \left(\frac{\sigma_M}{K\sigma_c} \right)^m \right] \quad (1)$$

In the low-stress regime ($\sigma_M \ll K\sigma_c$), the first-order Taylor expansion ($e^{-x} \approx 1 - x$ for $x \ll 1$) applied to the full Weibull distribution exactly recovers the simplified expression used in the model, $P_R = (\sigma_M/K\sigma_c)^m$. In the high-stress regime, although P_R can formally exceed unity, this does not affect the physical behavior: a value $P_R \geq 1$ simply guarantees removal of the element, which is exactly what the full Weibull distribution predicts in that limit. This approach, although not explicitly stated in the original text, is the same one adopted by Parkinson et al. in their original work.